

MEMORIA CIENTÍFICO-TÉCNICA PROYECTOS INDIVIDUALES
Convocatoria 2025 - «Proyectos de Generación de Conocimiento» y actuaciones para la formación de personal investigador predoctoral asociadas a dichos proyectos.

1. PROPOSAL DATA

PI 1 (Name and surname): Sara Barja Martínez (SB)

PI 2 (Name and surname): Matteo Ranaboldo (MR)

TITLE OF THE PROJECT (ACRONYM): Energy-Aware, Grid-resilient Location planning and flexiBility management for green data centres (**EAGLE**)

2. JUSTIFICATION AND NOVELTY OF THE PROPOSAL

2.1. Adequacy of characteristics and the purpose of selected modality.

This proposal aligns with the “research-oriented” (RO) modality, **Type A**. As defined in Annex III of the “State Plan for Scientific, Technical and Innovation Research 2024-2027”, the RO modality prioritizes contributions to key national and international challenges. Specifically, the **EAGLE** project directly addresses the **thematic priority** of “Energy and Mobility”, within the **strategic line** of “Energy Transition and Decarbonization.” Additionally, the **Type A** modality is intended for early-career researchers who have demonstrated technical expertise and the ability to tackle strategic societal challenges through applied research.

This proposal approaches data centres from an electrical engineering and energy systems perspective, complementing the traditional computational viewpoint. EAGLE views the data centre not as a data processing facility, but as a **large energy-intensive electrical load** with significant yet underused **flexibility potential**. Furthermore, to exploit the capacity of data centre (DC) installations for providing grid support, flexibility should be activated not only in energy-intensive hyperscale data centres but also across smaller and medium-sized data centres. The proposal focuses on two core areas: 1) **Optimal data centre location planning** based on grid topology, hosting capacity and congestion constraints, while optimizing sizing of on-site renewable and storage resources to increase self-sufficiency and reduce grid stress; and 2) **Flexibility and energy management strategies** for data centre electrical demand, with a focus on cooling systems and storage units as primary sources of flexibility. Both contribute to a more resilient power system, supporting energy transition and decarbonization goals. Last, but not least, EAGLE contributes to several United Nations Sustainable Development Goals (SDGs): SDG 7 (Affordable and Clean Energy) by prioritizing DC locations with high renewable-energy penetration; SDG 9 (Industry, Innovation and Infrastructure) by advancing tools for demand response in flexible DC loads such as cooling and storage systems and enabling ancillary services to the electrical grid when needed; and SDG 13 (Climate Action) by reducing the carbon footprint of data centres and assessing future climate risks scenarios in potential site locations to minimize future vulnerabilities in critical infrastructure.

2.2. Justification and expected contribution of the project to the generation of knowledge on the theme of the proposal. Starting hypothesis

2.2.1 Motivation

Data centres (DCs) constitute one of the fastest-growing electricity-intensive infrastructures globally. According to the International Energy Agency (IEA), DCs consumed approximately 1.5% of global electricity in 2024, with demand increasing at a rate of 12% annually over the last five years, driven by the proliferation of Artificial Intelligence (AI) and cloud services. Individual hyperscale facilities (massive and highly scalable data centres) are projected to reach power capacities of 100 MW, with annual consumption equivalent to 100 000 households [1]. Concurrently, the IEA also warns that grid congestion has become a major bottleneck to energy security and transition [2], posing significant challenges to the expected expansion of DC capacity [3]. In the European Union (EU), data centres consumed approximately 65 TWh (2.6%) of electricity in 2022 [4], with projections indicating a +150% increase by 2035 [5]. Despite this accelerating trend, the European data centre market grew at only half the global average rate in 2024, lagging behind global leaders such as the United States and China

[6]. To reverse this trend, the European Commission launched the “*AI Continent Action Plan*” in April 2025, setting a strategic objective to **triple data centre capacity** across the EU within five to seven years [7]. This initiative aims to strengthen Europe’s digital infrastructure and ensure technological sovereignty in strategic sectors, including AI, cloud services and high-performance computing. However, achieving this expansion target requires the active involvement of power system stakeholders and strict alignment with energy transition policies, such as Spain’s *Integrated National Energy and Climate Plan* (PNIEC) [8] and the European Green Deal [9]. These climate targets include reducing greenhouse gas (GHG) emissions, increasing the penetration of renewable energy in final energy use, and increasing the share of renewable electricity generation. **EAGLE** aims to bridge the gap between rising power consumption and climate targets, enabling sustainable expansion compliant with emission reduction strategies.

To secure sustainable growth within the data centre sector, the EU has established a legal framework through *Directive (EU) 2023/1791* (the recast Energy Efficiency Directive) [10]. Specifically, *Article 12* emphasizes the need for highly energy-efficient data centres, and it requires Member States to ensure that data centres with an installed Information Technology (IT) power demand of at least 500 kW publish the information detailed in *Annex VII*. Additionally, the Directive states that demand-side flexibility is a requirement for transitioning to a net-zero emission sector. As the penetration of renewable generation increases, the activation of such flexibility becomes essential [11]. Hence, integrating data centres, which are naturally energy-intensive loads, into demand-response mechanisms becomes a necessary strategy for supporting the stability and management of the electrical grid. In this context, energy-efficient data centres are a necessity for addressing the environmental implications of our digital age. Consequently, the industry must shift toward “**green data centres**”, i.e. facilities that aim to maximize efficiency and minimize environmental impact throughout their lifecycle using advanced technologies. This includes optimized cooling systems, renewable energy integration, waste heat recovery, high-efficiency hardware, and sustainable building design. Life Cycle Assessment (LCA) is employed to evaluate the environmental impacts caused by an electrical generation source throughout its lifetime, from the extraction of raw materials to disposal or recycling (known as cradle-to-grave) [12]. In the context of the electricity sector, the Global Warming Potential (GWP) indicator quantifies the GHG emissions per kWh across various generation sources, including battery storage systems.

This landscape serves as a key driver for this proposal. Compliance with efficiency and reporting requirements necessitates a significant advancement in the state of the art, specifically regarding the tools available for **sustainable data centre long-term planning and short-term energy management**. Based on this analysis of the current context and the critical motivations driving this proposal, the integration and expansion of DC loads into a constrained energy system present **four major challenges**:

CH1) Mitigating grid congestion. Overcoming constraints in grid access in saturated network areas, caused by increasing electricity demand.

CH2) Optimizing renewable integration. Identifying optimal regions for large-scale on-site integration of renewable generation and storage for DCs.

CH3) Unlocking demand-side flexibility. Enabling DCs' demand-side flexibility through intelligent energy management to support grid stability and reduce costs and emissions.

CH4) Ensure data centre sustainability. Addressing CH1–CH3 while ensuring national and international decarbonization targets [8], and social welfare.

EAGLE addresses these challenges by covering the entire lifecycle of data centre expansion, ranging from long-term spatial planning and site selection to the techno-economic sizing of on-site renewable energy systems and short-term flexible operation. Throughout each phase, the project integrates environmental, economic, and social criteria to ensure a net positive impact at both regional and national levels. These challenges will be addressed by EAGLE project through optimization models, including multi-criteria and multi-objective formulations, as well as AI-based algorithms for forecasting and managing data centre energy consumption. These tools will be integrated into a unified framework that covers both long-term planning and short-term (day-ahead) operation, thereby addressing the four identified challenges.

2.2.2 State-of-the-art and beyond

In this subsection, we outline the research gaps that this project aims to address, defining the current state of the art related to the problem and explaining our proposed contribution to advance beyond it and further develop the field.

Optimal location planning of data centres

State-of-the-art. Data centre location planning has evolved into a multidimensional optimization challenge that must balance various aspects, including grid stability, sustainability objectives, and climate resilience. Also, a proper DC location planning needs to consider **grid capacity** (in order to minimize congestion problems and fulfil flexibility requirements), **expected cooling demand** (which can represent up to 40% of the DC total consumption [13] [14]), **available renewable resources** (in order to maximize DC autarky and minimize its impact on the grid), and **environmental risks**, which are all site dependent. To achieve this, a comprehensive multicriteria planning process is required in which the interaction and knowledge sharing among stakeholders are crucial [15] [16]. On the other hand, today's location planning is still mostly an isolated process carried out by single entities, which are often lacking a comprehensive view, therefore missing sustainability opportunities and harming grid resilience [16]. In this context, existing data centre location tools (such as *ENTSO-E* [17], *dcBlox* [18] and *451 Research* [19]) tend to prioritize **single-objective optimization** (e.g., minimizing energy costs or carbon intensity), revealing a critical absence of holistic, multi-criteria decision-making frameworks [16]. Some tools focus on optimizing the cooling system [13], others on grid capacity expansion and generation planning (*PyPSA* [20]), but there is no tool that explicitly considers most grid and energy-related aspects, such as grid constraints, expected consumption and renewable resources, at the same time [16] [21]. Additionally, climate risk is generally not considered in planning, as more than 40% of the top 100 data centre hubs are located in areas at high or very high risk of climate change [22]. In fact, most available tools utilise static datasets (annual average temperatures) based on historical data (ASHRAE guidelines [23]), but fail to account for projected temperature rises, increased extreme weather events, and sea-level rise, which threaten coastal DCs. It is also important to highlight that many tools, such as *ENTSO-E* [17] are biased toward hyperscale DCs, with huge capacities of more than 50 MW, such as Google, AWS, Microsoft, and neglect edge/micro DCs, with small capacities lower than 100 kW, which have unique constraints [23]. In this sense, most proprietary tools such as *dcBlox* [18] and *451 Research* [19] are **expensive and restricted to large corporations**, limiting adoption by small and medium enterprises (SMEs) or public-sector (e.g., government), and open-source alternatives (e.g., *OpenDCIM* [24], *NetBox* [25], *OpenEnergyTransition* [26]) lack sophisticated energy modules or are not focused on DC. In particular, there is no open-source specifically designed for the location planning of DC that can be used by European public administrations.

Beyond the state-of-the-art. As stated, proper DC location planning is a complex task that requires considering multiple site-dependent aspects, including grid, energy, and climate-related factors, to enhance overall DC sustainability and support grid stability and resilience. However, a planning framework to support DC location that relies on a comprehensive view of all these aspects is still missing among grid planners and public institutions. In this context, one of the aims of this project is to develop a multicriteria optimization algorithm (named **SiteOptimizer**) that combines grid constraints, electricity demand, on-site renewable resources, and environmental constraints to identify the most appropriate DC location(s). A **multi-criteria** optimization method [27] will be developed in order to properly balance all the different objectives, including **economic, environmental, grid-impact** and **social criteria**. The SiteOptimizer will also consider environmental constraints, such as **climate risk** data, e.g. flooding and wildfires probability maps [28], and access to water, particularly relevant in countries like Spain. In this sense, to address uncertainties in future scenarios, SiteOptimizer will enable performing sensitivity analyses to simulate a wide range of scenarios e.g. climatic scenarios, by adjusting key input parameters in the optimization model. SiteOptimizer will rely on publicly available databases, making the most of recent European and global open-source initiatives to access high-quality data on, for instance, renewable resources [29] [30], grid data [31] [32] [26] and climatic projections [33]. By taking into account the real expected demand of each specific DC, the tool will be flexible enough to support the planning of **DC of different sizes** (including edge and micro-DC, which are currently overlooked by existing tools). Furthermore, it will

be specially designed to be used by the public sector and small and medium enterprises that are currently missing such support. In this sense, the multicriteria optimization model will be integrated into an **open-source** Geographical Information System (GIS) interface to support grid planners, SMEs, and public institutions in properly locating green DCs and planning the DC capacity evolution over the next few years. This should enable the analysis of all relevant information with transparency, thereby facilitating **the interdisciplinary decision-making process** that is much needed in this field. The SiteOptimizer proposal builds upon optimization models developments and energy planning experience of both Principal Investigators (PI), in particular, **PI2-MR** has been working with optimization methods for energy systems planning for more than 10 years [34] [35] [36], while **PI1-SB** has also experience in multi-criteria optimization [27] as a core topic of her doctoral thesis.

Green data centres for emission reduction and renewable integration

State-of-the-art. Data centres are increasingly identified as a major source of GHG emissions, recently ranked as the second-largest contributor within the Information and Communication Technology (ICT) sector [37]. As an example of this trend, Google reported a 17% increase in electricity consumption of its data centres between 2022 and 2023 and a 50% rise in emissions over the past five years. Consequently, the company set in 2025 an ambitious target to operate on 24/7 carbon-free energy by 2030 [38]. However, validating the achievement of this target remains a challenge. Currently, data centres rely on **annual volumetric matching** [39], a mechanism that allows operators to claim *net-zero* status even while consuming fossil-fuel-heavy grid power during periods of low renewable generation. Recent analyses have demonstrated that annual net-zero approaches often fail to achieve true carbon neutrality when assessed against dynamic hourly fuel mixes [39] [37]. Consequently, the field is transitioning toward **hourly carbon-free energy** (CFE) scores. Achieving high CFE scores requires a diversified energy portfolio that combines variable renewables with clean firm power and long-duration energy storage. Furthermore, the integration of demand response mechanisms is essential to align DC demand with clean energy availability [40], as also stated by the EU Directive [10]. Current frameworks for DC planning, sizing, and operation fail to incorporate LCA and critical *non-carbon* environmental impacts such as water scarcity, land-use change, which are in line with the main objectives of the European Green Deal [9]. While widely used tools like HOMER [41], and REopt [42] effectively dimension on-site resources (renewables, storage, backup) based on economic optimization and resilience constraints, they have limitations. Specifically, these platforms lack the geospatial planning required for optimal site selection. Furthermore, they ignore the embodied carbon of energy assets and the broader ecological footprint, thereby obscuring the true environmental impact of the facility. Sustainability must be a key component in all modern developments, especially when dealing with energy-intensive infrastructures, such as DCs. Thus, the reduction of the environmental impact must be embedded within all the different development phases, from planning and design to the operation of “green data centres”. However, most existing tools that support DC design and operation are focused on minimizing costs or are focused on specific energy or carbon metrics, such as the Power Usage Effectiveness (PUE).

Beyond the state-of-the-art. In this proposal, all the developments will include a sustainable approach and are focused on minimizing the environmental impact (over the whole life-cycle, defining new metrics if needed) of the DC, both in the long-term planning and in the short-term energy-management. Regarding **planning**, besides specific environmental considerations already mentioned about location planning (**SiteOptimizer**), this proposal introduces a comprehensive method for sizing renewable energy systems and storage to enhance DC **self-sufficiency** (named **OptiSize**). Unlike static methods based on annual volumetric matching, the method enforces a high hourly carbon-free energy score that reflects true carbon impact, dimensioning a hybrid mix of on-site renewables, clean baseload power, a variety types of storage, and backup systems like photo-voltaic arrays, wind energy and hydrogen fuel cell systems to supplement or partially replace conventional diesel backup. Where on-site generation is physically limited to cover all DC demand, the framework will suggest the amount of capacity to be covered with off-site Power Purchase Agreements (PPAs). OptiSize will employ a weighted multi-objective function, allowing operators to prioritize **economic criteria** (minimizing LCOE/CAPEX) or **environmental criteria** (minimizing LCA impacts including emissions, land-use and water usage, based on cradle-to-grave LCA of all energy resources) depending on specific project

constraints. Regarding **energy management**, the DC Energy Management System developed in this project (named **DC-EMS**) integrates **life-cycle assessment** (LCA) components directly into daily operations. By actively managing flexibility in cooling systems and storage units, the system maximizes self-consumption while reducing consumption-related GHG emissions, strictly meeting resilience constraints. Building on prior research by **PI1-SB** [27] [43], which successfully demonstrated LCA-based flexibility in domestic energy systems, this proposal integrates comprehensive LCA indicators directly into the planning, sizing, and short-term energy management of green data centres. Parallely, bringing on-site renewable generation (field in which **PI2-MR** has extensive experience [44] [45]) will facilitate data centres integration in the power network, offering a roadmap for power systems decarbonization.

Demand-side flexibility at data centres

State-of-the-art. Data centres typically operate as passive baseload consumers, inflexible to grid conditions. However, given their high energy intensity, they offer significant potential for flexibility activation. While European regulations increasingly emphasize the need for demand-side flexibility to support grid stability [7] [9], still further legislative development is required to fully facilitate DC participation in energy markets and ancillary services [21]. Recent literature identifies three primary sources of flexibility within a DC: **electrical assets** (uninterruptible power supplies (UPS), batteries, generators), **thermal assets** (cooling systems, waste heat recovery), and **computational loads** [21]. In this proposal, the first two are considered. **Electrical flexibility:** Beyond battery storage, UPS represent a potential flexible resource. Studies indicate that 10–50% of deployed UPS capacity is surplus energy not required even during outages [21]. An advanced flexibility energy management system can activate this surplus for ancillary services while maintaining the minimum capacity required for reliability and power outages. **Thermal flexibility:** Cooling demand fluctuates based on ambient temperatures and IT utilization. Strategies such as precooling, setpoint adjustment, and thermal storage allow for "Dynamic Cooling Management." This approach enables DCs to overcool during periods of off-peak demand or high renewable availability, thereby storing thermal inertia to reduce consumption during peak periods [46] [21]. However, such strategies must be strictly constrained by the ASHRAE TC 9.9 guidelines [47], which define the operational temperature and humidity ranges necessary to ensure equipment reliability. Despite these technical possibilities, significant gaps remain in the state-of-the-art. Leading commercial tools (such as *Spacewell*, Schneider Electric's *EcoStruxure*) prioritize real-time monitoring of power and temperature rather than active flexibility optimization. In addition, a recent study confirms that while flexibility consistently reduces operational costs, it may inadvertently increase emissions depending on the local grid mix [48]. Consequently, optimization based only on cost is insufficient for ensuring sustainable DCs.

Beyond the state-of-the-art. To address these limitations, the EAGLE project proposes a **multi-objective energy management system (DC-EMS)**. This framework optimizes cooling infrastructure, battery storage and UPS by balancing cost reduction with carbon footprint minimization, accounting for on-site renewable generation and real-time grid GHG emissions. The DC-EMS can also attend **flexibility requests** from Transmission System Operators (TSOs) and Distribution System Operators (DSOs) for grid support. The flexibility of IT power consumption is excluded from this scope, as the scheduling of computational workloads requires distinct domain expertise. This proposal builds upon validated demand-side models. **PI2-MR** previously led the development of production management models [49] [50] for enabling **industrial demand response** [11] for the European project FLEX4FACT [51], while **PI1-SB** possesses extensive experience in energy management systems for the residential and community sectors, as stated in her PhD. Thesis [52]. Also, both PIs have experience in developing forecast algorithms for energy prices [53] and renewable generation [54]. These combined competencies provide a strong technical baseline for the EAGLE framework. The EAGLE project advances demand flexibility by deploying an advanced DC-EMS that **unlocks the flexibility potential of data centres** through the unified control of cooling, on-site renewables, and battery storage. By developing novel algorithms for dynamic cooling, the DC-EMS optimizes energy use while strictly maintaining equipment health, utilizing a multi-criteria objective function that dynamically prioritizes cost or GHG emission reductions on hourly basis. EAGLE expands the scope of flexibility by leveraging UPS and backup storage units to provide ancillary services and rapid congestion relief, thereby enhancing grid resilience without the need for immediate infrastructure expansion.

2.2.3 Project contributions

This project addresses the challenges identified in Section 2.2.1 through three key contributions:

- (1) **Long-term location planning framework** based on a multicriteria optimization to identify optimal locations for new data centres, considering climate resilience, renewable potential, hosting capacity and grid constraints, leveraging recent European and global open-source initiatives to access high-quality public datasets
- (2) Optimal **techno-economic sizing of on-site renewable generation and energy storage** systems to ensure a minimum level of self-sufficiency, reducing peak demand and grid congestion
- (3) **Short-term flexibility and energy management** system to optimize the operation of data centre cooling systems, enabling demand-side flexibility and prioritizing self-consumption of on-site photovoltaics (PV) and storage.

2.2.4 Project starting hypothesis

Based on the state of the art and identified research gaps, the following hypotheses guide this project:

H1) The expansion of data centres is increasingly constrained by grid capacity limitations, which cause deployment delays and, in some cases, block deployments. This bottleneck will likely intensify in the coming years due to the growing demand driven by cloud computing, AI and edge services [2], [3].

H2) Optimal data-centre location that simultaneously considers multiple site-dependent grid hosting capacity, local climate conditions and renewable energy availability can significantly reduce DC connection delays, minimize the need for capital-intensive grid reinforcements, and maximize on-site renewable resources utilization [15] [16].

H3) Uninterruptible power supplies systems, including batteries and backup generators, can be effectively dimensioned and operated not only for reliability but to enhance data centres self-sufficiency and provide active flexibility services to the grid [21].

H4) The thermal inertia inherent in data centre cooling systems can function as *virtual energy storage*, enabling significant demand-response capabilities without compromising IT equipment performance or reliability [13].

H5) To operate as a flexible grid load, data centres require an integrated energy management system capable of optimally coordinating their flexible assets and on-site generation to respond to dynamic electricity prices, provide ancillary-service requests for grid stability support and reduce GHG emissions [55].

In research-oriented projects

2.3. **Justification and expected contribution of the project to solving specific problems linked to the selected thematic priority**

The proposed project is aligned with the strategic objectives of Spain's *National Integrated Energy and Climate Plan 2021–2030* [8], particularly in the areas of **energy transition**, **digitalization**, and **power system decarbonization**. Within the thematic priorities of the *Plan Estatal De Investigación Científica, Técnica y de Innovación 2024-2027* (PEICTI), this project contributes directly to the "**Energy and mobility**" thematic priority, specifically addressing the strategic line "**Energy transition and decarbonization.**" More precisely, it focuses on the following areas of intervention: "*Renewable energy for thermal and electrical use and energy storage*", "*Smart grids for advanced renewable energy management*" and "*energy efficiency*". In particular, EAGLE will contribute to achieving the following objectives established in the PNIEC 2023–2030 [8]:

- (i) "**32% reduction in greenhouse gas emissions compared to 1990**": EAGLE integrates LCA methodology across three data centre key phases: optimal location planning, on-site renewables and storage sizing, and short-term energy management. By embedding environmental constraints directly into the objective functions, the project ensures a robust method of carbon footprint accounting and reduction.
- (ii) "**48 % renewables in final energy consumption**": (iii) "**81 % renewable energy in electricity generation (..)** To enable this, various factors that will provide flexibility to the electricity system are important, such as energy storage, demand management(...)": The project optimizes data

centre design to maximize on-site renewables and self-consumption by analyzing regional renewable resources and identifying optimal locations for on-site renewables. Also, in short-term operation, DC-EMS maximizes the utilization of this renewable generated energy by integrating also battery storage and activating flexibility in assets like cooling systems.

- (iv) “Reduction of energy dependence by up to 50 %”: By maximizing on-site renewable self-consumption and utilizing storage to match demand with generation, EAGLE reduces its dependence on external energy sources.

3. OBJECTIVES, METHODOLOGY AND WORK PLAN

3.1. General and specific objectives.

The EAGLE project aims to align the rapid growth of data centres with EU climate goals by creating a unified framework for long-term sustainable planning and short-term operational flexibility. By integrating a multi-criteria site assessment that covers grid hosting capacity, renewables, and environmental constraints, along with an advanced energy management system, the project delivers a service to guide optimal low-carbon data centre infrastructure deployment and operation. Table 1 presents the project’s main and specific objectives.

ID	Objective	WP/Task
O1	Identify inputs and constraints for planning and operation of green data centres	WP2
O1.1	Obtain DC load profiles by disaggregating IT and cooling demand to quantify the flexibility potential.	T2.1
O1.2	Evaluate on-site renewable energy resources and storage technologies suitable for green data centres.	T2.2
O1.3	Map nodal grid hosting capacity and environmental constraints to establish geospatial boundary conditions for optimal site location.	T2.3
O2	Develop a long-term planning framework for green data centres	WP3
O2.1	Develop a multi-criteria spatial optimization model to identify optimal DC locations considering grid, energy, climate, environmental and social constraints (<i>SiteOptimizer</i>)	T3.2
O2.2	Create an optimal sizing algorithm for on-site generation and storage assets (<i>OptiSize</i>)	T3.1
O2.3	Implement an open-source, GIS-based visualization interface to display optimal data centre location and relevant site-dependent inputs.	T3.2
O3	Design short-term operational control strategies for energy management and grid flexibility provision	WP4
O3.1	Develop forecasting models for day-ahead IT workload, cooling demand, energy prices and on-site renewable generation profiles.	T4.1
O3.2	Develop a mathematical thermal model of the cooling system to control thermal inertia for short-term flexibility.	T4.1
O3.3	Design an Energy Management System (<i>DC-EMS</i>) to minimize electricity cost and GHG emissions	T4.2
O3.4	Develop the framework to enable DCs participation in ancillary service markets.	T4.2
O4	Validate the proposed algorithms through scenario analysis and quantify impact	WP3, WP4
O4.1	Conduct long-term scenarios and sensitivity analysis to stress-test the resilience, environmental impact and economic viability of the proposed solutions.	T3.3, T4.3

Table 1 General and specific objectives

3.2. Description of the methodology.

The methodology to achieve the objectives identified in Section 3.1 includes the following strategies:

- **S1. State-of-the-art Analysis.** A comprehensive review of data-driven approaches for optimal location planning and demand-side flexibility in data centres. This includes an in-depth analysis of grid constraints, renewable and storage integration, and demand response possibilities. Furthermore, existing cooling optimization models and storage technologies will be evaluated with a specific focus on their flexibility potential.
- **S2. Data acquisition and Preprocessing.** Collection of input data for planning and operation algorithms, primarily relying on publicly available datasets. Where specific data is unavailable, synthetic data will be generated. All datasets will be cleaned, normalized, and post-processed to ensure high-quality inputs for algorithm development and testing.

- **S3. Algorithms development.** Design, development, testing, and evaluation of algorithms for optimal location planning (*SiteOptimizer*), renewable and storage sizing (*OptiSize*), and energy management for activating flexibility and providing ancillary services when needed (*DC-EMS*). This step integrates the findings from S1 and utilizes the input data prepared in S2.
- **S4. Sensitivity analysis.** Execution of sensitivity analyses to evaluate and compare the performance of the algorithms developed in S3 across multiple scenarios.

3.3. Work plan and schedule.

The project will be developed through the following Work Packages (WPs):

- **WP1:** Project management, communication and dissemination
- **WP2:** Data acquisition and preparation (see Objective **O1**)
- **WP3:** Long-term planning of green data centres (see Objective **O2, O4**)
- **WP4:** Short-term flexibility and energy management of green data centres (see Objectives **O3, O4**)

Figure 1 illustrates the conceptual framework of the EAGLE project and the interdependencies between the tasks (T). The Gantt chart is presented in Table 3 details the project timeline, including task durations, and milestones. **WP1** is transversal to the project. **WP2** establishes the general framework and provides the data inputs required for the algorithm development and assessment activities carried out in **WP3** and **WP4**, which run in parallel. The content of each WP is detailed in its corresponding Tasks (T), describing the scope, methodology, and innovative aspects.

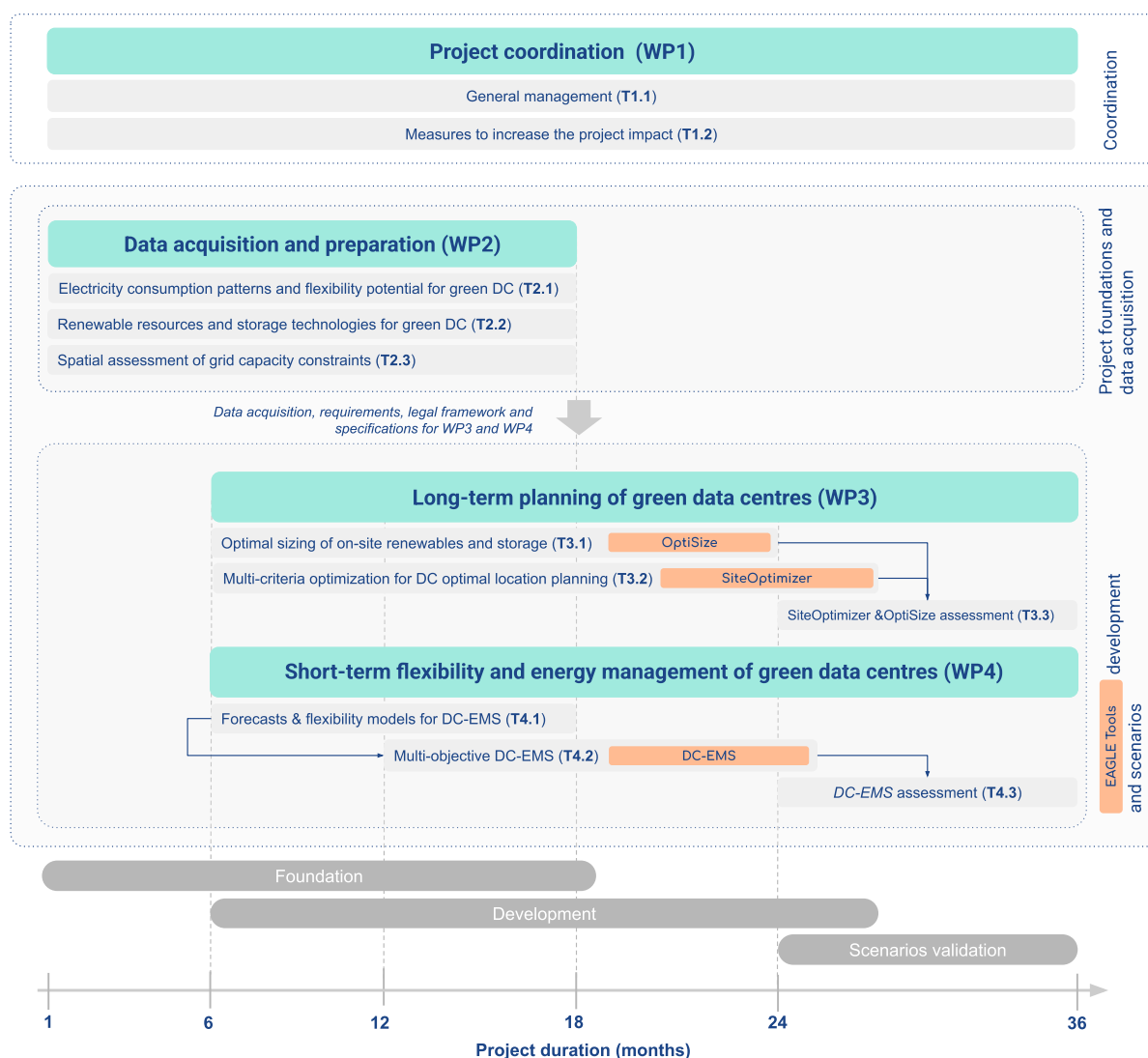


Figure 1 EAGLE Framework. WPs and Tasks interaction

For each Task, the start-end month (M) are specified, along with the responsible persons, participants, expected results, and milestones (MS).

WP1. Project management, communication and dissemination

Responsible: SB, MR **Duration:** M1-M36

This Work Package (WP) ensures the project goals are achieved efficiently within the allocated time and budget. It orchestrates the administrative and technical coordination (T1.1) while maximizing the project's impact through a robust *Dissemination and Communication Plan* (T1.2). The main objectives of WP1 are (i) **Coordination**. To implement an Agile/Scrum methodology for resource optimization, quality control, and risk mitigation (ii) **Reporting**: provide effective reporting and communication within the project team and towards the Ministry; (iii) **Impact**: to execute a multi-channel strategy that transfers knowledge to the scientific community, industry, and the general public ("*Science to Society*").

T1.1 General management			M1-M36
The team will develop a <i>Project and Data Management Plan</i> (PDMP) at the beginning of the project (M6), which will be revised and updated through follow-up project evaluation meetings. The team will implement an Agile/Scrum methodology to ensure the project's dynamic coordination. Workflows will be organized in weekly sprints using Trello (or similar) for real-time task tracking, early bottleneck detection, and seamless iteration. A cloud repository for the project (Google Drive) will ensure secure document versioning. Bi-annual project evaluations will be conducted by the responsible and team members to validate milestones, assess risks, and ensure alignment with the PDMP; if needed, updates will be made accordingly. Technical reports will be generated for the Ministry.			
Responsible	Participants	Expected results	Milestones
SB, MR	SG, AS, PG	PDMP; Meeting Minutes	MS1
T1.2 Measures to increase the project impact			M1-M36
This task deploys a comprehensive strategy for the project dissemination and communication goals and outcomes, targeting three main audiences: academic, industrial, and general society (" <i>Science to Society</i> "). The different channels described in Section 4.4 will be used to broaden the reach of the project. Results will be published in high-impact open-access peer-reviewed journals and presented at key international conferences in the energy field and webinars. To raise awareness among the general public, we will use the University media office for press releases, keep an active project website, and use social media (LinkedIn) to explain the project's societal benefits. A dedicated <i>Knowledge Transfer Workshop</i> will be organized with industrial partners and public administration entities to explore exploitation pathways and practical applications of the findings.			
Responsible	Participants	Expected results	Milestones
SB, MR	SG, AS, PG	Scientific publications for dissemination; Non-scientific publications for communication; Presentations; Events attendance/organization; Networking with potential stakeholders	MS1

WP2. Data acquisition and preparation

Responsible: SB, MR **Duration:** M1-M18

WP2 focuses on defining and preparing the most relevant technical inputs for planning and energy and flexibility management of green data centres. In particular, the electricity consumption and its flexibility potential (T2.1), the on-site renewable potential and storage technologies (T2.2), and the grid constraints (T2.3) are accessed. The results of these tasks will serve as main inputs for the algorithms to be developed in WP3 (T3.1 and T3.2) and WP4 (T4.1 and T4.2).

T2.1 Electricity consumption and flexibility potential for green DC	M1-M18
This task focuses on assessing the main patterns and drivers of the two main sources of electricity consumption in data centres: cooling and data processing (IT). Regarding cooling, the state-of-the-art of the different types of cooling technologies currently in use for DC and their cooling requirements will be reviewed [55], in order to define a methodology for estimating the cooling expected	

consumption depending on the cooling technology and climate conditions. Regarding processing and computation consumption, the assessment will be based on publicly available datasets, in order to evaluate its consumption patterns and estimate the flexibility potential. From available datasets, such as Alibaba [56] or other data centre datasets [57]), additional consumption scenarios can be generated if needed. To ensure high-quality data, the Detection of Measurement Errors (DOME) tool developed by PI1-SB will be used to clean and validate the datasets, enhancing the performance of the algorithms implemented in WP3 and WP4. Current regulation about flexibility requirements for DC will also be reviewed.

Responsible	Participants	Expected results	Milestones
SB	AC, PG	Identification of main open-sources datasets about DC consumption; Review of cooling technologies for DC	MS2
T2.2 Renewable resources and storage technologies for green DC			M1-M18
This task will assess the renewable (in specific wind and solar) potential and define proper storage technologies to reduce data centres dependency from the electric grid and offer flexibility. For the renewable potential, most recent available databases [30] will be reviewed to estimate wind and solar resources. Regarding technology, particular focus will be given on analysing existing storage technologies to select the most appropriate ones for providing flexibility services by green DC.			
Responsible	Participants	Expected results	Milestones
MR, SB	SG, TE, AS	Identification of main open-sources datasets about renewable energy and DC consumption; Review of storage technologies for DC	MS2
T2.3 Spatial assessment of grid capacity constraints			M1-M18
This task will assess the grid capacity constraints at various locations in order to support the planning and energy management of DC. The analysis will identify critical power system parameters, such as line saturation levels and regional flexibility requirements. To reconstruct the High Voltage (HV) network topology, we will examine the most recent public datasets and open-source European grid initiatives, such as the <i>PyPSA-Eur</i> open-source platform [31], <i>MapYourGrid</i> initiative [32], the <i>Open Energy Transition project</i> [26], and the <i>European GridKit Zenodo</i> repository [58], incorporating high-voltage and, where available, medium-voltage data.			
Responsible	Participants	Expected results	Milestones
MR	SG, AS, PG	Identification of critical grid constraints sources for DC planning and flex management	MS2

WP3. Long-term planning of green data centres

Responsible: MR **Duration:** M6-M36

WP3 focuses on developing algorithms to support the long-term planning of data centres considering energy requirement, on-site generation, grid and environmental constraints. Relying on the assessments done in WP2, an optimal techno-economic sizing method for DC on-site renewable generation and storage will be defined (T3.1) and a multi-criteria optimization model will be developed to evaluate whether a certain location is appropriate for installing a DC considering technical, environmental and social constraints (T3.2). The techno-economic sizing method and the multi-criteria optimization model will be used to analyse a wide range of future scenarios (T3.3).

T3.1 Optimal sizing for on-site renewable energy and storage systems (*OptiSize*) M6-M24

This task will define a methodology for designing DC on-site generation and storage to maximize their self-sufficiency (*OptiSize*). Given the data centres estimated consumption (T2.1) and the renewable resource and technology (T2.2), the method will design the **optimal configurations for integrating renewable energy resources and storage** to enhance self-consumption and reduce grid congestion. The sizing optimization algorithm will consider multiple criteria in order to balance environmental, e.g. minimizing life-cycle impact, and economic sustainability, e.g. minimizing CAPEX or payback. The outputs of this model in term on-site generation and storage capacity will be used by the short-term

energy management algorithms (WP4), also to refine the results of the multi-criteria optimization (T3.2).

Responsible	Participants	Expected results	Milestones
MR, SB	SG, TE, AS	OptiSize model	MS3

T3.2 Multi-criteria optimization for DC optimal location planning (*SiteOptimizer*) M6-M28

In this task, a multi-criteria optimization model (*SiteOptimizer*) will be developed to evaluate whether a certain location is appropriate for installing a DC of a certain size, considering the expected **cooling requirements** (T2.1), the **on-site renewable potential** (T2.2), the **electric grid constraints** (T2.3). Besides those technical constraints defined in WP2, the model will be designed to include also **environmental constraints**, such as water access, land usage or climate risk data (e.g. European Climate Risk and Adaptation Platform [59], the probability of extreme events [60]), and social data. The SiteOptimizer will combine all this information, balancing economic (e.g., reducing cooling requirements), environmental (e.g., increasing on-site self-consumption), grid (e.g., reducing annual congestion probability), and social criteria to evaluate the adequacy of a certain location. The multi-criteria weights can be adjusted by decision-makers to evaluate the pros and cons of different options, thereby enabling a **transparent and intersectoral decision-making process**. The model will be validated with case studies in Spain and Europe. An **open-source graphic user interface** will be developed to show the SiteOptimizer results on a regional/national level. Open-source user interface tools, such as Streamlit, will be used to display the SiteOptimizer results as well as the input data for the various layers included in the analysis (e.g., renewable resources, grid constraints, cooling demand).

Responsible	Participants	Expected results	Milestones
MR, SB	AS, TE	SiteOptimizer model	MS3

T3.3 *SiteOptimizer* and *OptiSize* assessment M24-M36

This task will focus on analysing the impact of different DC planning strategies in multiple scenarios. The OptiSize (T3.1) and SiteOptimizer (T3.2) will be used to simulate a wide range of “what-if” scenarios by adjusting key input parameters in the optimization models. These parameters may include grid constraints, climate risks, land availability, and social or environmental factors. For instance, different **future climate scenarios** can be analysed using the climatic projections (such as the recent Destination Earth projections [61]). SiteOptimizer results will also be compared with a baseline case (e.g. current DC locations). Specific LCA indicators will be used to compare strategies in different scenarios. This scenario-based approach allows stakeholders to explore discrete possibilities and gain valuable insights into how different conditions may influence long-term planning. This assessment promotes more **resilient and evidence-based decision-making** for the strategic expansion of data centre infrastructure.

Responsible	Participants	Expected results	Milestones
MR	AS, FT, TE	Quantification of improvements achieved by SiteOptimizer and OptiSize; Future climate scenarios analysis	MS5

WP4. Short-term flexibility and energy management of green data centres

Responsible: SB **Duration:** M6-M36

The primary objective of WP4 is to develop a multi-objective energy management system that optimizes data centre operations by leveraging flexible resources, specifically battery storage, thermal cooling inertia, and on-site renewable generation. The work is structured into three tasks: T4.1 develops the necessary forecasting algorithms and the thermal flexibility mathematical model; T4.2 builds the core DC-EMS optimization engine to balance electricity costs against GHG emissions and establishes the methodology for providing flexibility to grid ancillary services; and T4.3 assesses the DC-EMS impact across a set of regional and long-term climate-based scenarios.

T4.1 Forecasts and flexibility models for DC-EMS M6-M18

This task provides the essential input parameters required for the DC-EMS (Task 4.2). It develops day-ahead forecasting models for on-site renewable generation, data centre's computational (IT) load and

cooling electrical demand, using the datasets collected in WP2. Additionally, a **mathematical thermal model** of the cooling system will be formulated based on the flexibility patterns defined in Task 2.1 and a **comprehensive literature review of optimal thermal control strategies for DCs**. This model will quantify the system's thermal inertia to characterize cooling demand as a flexible resource. The resulting model will be integrated as a new module within the DC-EMS framework (Task 4.2) and validated using publicly available DC datasets.

Responsible	Participants	Expected results	Milestones
SB, MR	AC, PG	Day-ahead AI-based forecasting models; Thermal mathematical model for DC-EMS	MS4

T4.2 Multi-objective DC-EMS (*DC-EMS*)

M12-M26

Building upon inputs from T4.1, this task develops the **DC-EMS** model to optimize the day-ahead operation of the data centre. The DC-EMS will use a **multi-objective function** designed to minimize both electricity costs and environmental impact, explicitly accounting for GHG and LCA-based emission factors associated with on-site renewables, storage units' usage, and the time-varying grid energy mix. A dynamic weighting parameter will be implemented to calibrate the objective function, allowing DC operators to prioritize either costs or emissions based on regulatory constraints or operational goals. Furthermore, this task defines **optimal strategies for activating flexibility resources** (such as batteries, UPS and cooling systems) in response to TSO/DSO flexibility requests. This includes developing the specific protocols required for the DC-EMS to provide ancillary services and support grid stability.

Responsible	Participants	Expected results	Milestones
SB	AC, PG	Short-term operational DC-EMS optimization algorithm; framework for flexibility activation for ancillary services.	MS4

T4.3 DC-EMS assessment: regional and long-term scenarios

M24-M36

This task evaluates the performance of the DC-EMS across defined scenarios to quantify the data centre potential in terms of economics, environmental impact, and flexibility. Key performance indicators (KPIs) will be calculated to compare two specific scenario categories. First, **(i) Regional flexibility assessment** compares the performance of a selected green DC using the DC-EMS across diverse geographical regions over a representative period (i.e. five years). T3.2 outcomes will identify the regions that might be most suitable for performing these scenarios. Each region has different outdoor temperatures and varying availability of renewable energy, which influences the performance of DC-EMS. Cooling demands and generation forecasting models will be run with each local weather data to run the scenarios. Open historical and hourly weather data is available at [29] [30]. This analysis identifies regions that offer superior flexibility potential, higher self-consumption rates, and reduced GHG emissions. Second, **(ii) Long-term climate resilience** scenarios perform sensitivity analyses to simulate the impact that climate change might have on DC electricity bill costs, emissions and flexibility activation. While the methodology is the same as the one followed in the regional assessment (i), this scenario category stress-tests the DC-EMS by deliberately modifying weather input data. Specifically, historical weather conditions, such as temperature profiles, will be synthetically adjusted (e.g., increasing by +2°C to +5°C) consistent with climate projections, while maintaining the DC's technical characteristics constant. Results from the regional assessment will serve as a baseline to quantify how extreme climate conditions over the coming decades may impact DC performance, emissions, costs and cooling requirements.

Responsible	Participants	Expected results	Milestones
SB	AC, PG, TE, FT	regional flexibility potential and economic/environmental performance; long-term sensitivity analysis on the impact of future climate change on DC operations	MS5

3.3.2 Summary of milestones and schedule.

The project execution is monitored through the five milestones (MS) depicted in Table 2 to ensure the achievement of the general and specific objectives (Section 3.1, Table 1):

Id	Title	Due date
MS1	Project management, dissemination and data management plan established	M6
MS2	Data acquisition and technical knowledge base has been completed	M18
MS3	Long-term optimal sizing and location planning algorithms (OptiSize and SiteOptimizer) have been developed and validated	M24, M32
MS4	Short-term DC-EMS algorithms and day-ahead forecasting models have been developed and validated	M18, M32
MS5	Final assessment of multiple scenarios has been completed	M36

Table 2 EAGLE Milestones throughout the project duration

The EAGLE project duration is 36 months (3 years). The temporal distribution of WPs, tasks, and milestones is illustrated in the Gantt chart detailed in Table 3.

	Month (M)	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36
WP1 Project management, communication and dissemination																			
T1.1 General management				MS1						MS1						MS1			
T1.2 Measures to increase the project impact				MS1						MS1						MS1			
WP2 Data acquisition and preparation																			
T2.1 Electricity consumption and flexibility potential for green DC										MS2									
T2.2 Renewable resources and storage technologies for green DC										MS2									
T2.3 Spatial assessment of grid capacity constraints										MS2									
WP3 Long-term planning of green data centers																			
T3.1 Optimal sizing for on-site renewable energy and storage systems													MS3						
T3.2 Multi-criteria optimization for DC optimal location planning															MS3				
T3.3 SiteOptimizer and OptiSize assessment																			MS5
WP4 Short-term flexibility management of green data centers																			
T4.1 Forecasts and flexibility models for DC-EMS										MS4									
T4.2 Multi-objective DC-EMS														MS4					
T4.3 DC-EMS assessment																			MS5

Table 3 EAGLE project Gantt chart

3.4. Identification of critical points and contingency plan.

The identification of possible risks, corresponding mitigation measures, likelihood of occurrence and its expected impact on the project are summarized in Table 4.

Risk/ challenge	WP	Mitigation measures	Likelihood	Impact
Delays in recruitment or lack of funding for Technical Engineer (TE)	1	(i) Launch recruitment calls after project acceptance via EURAXESS/ UPC networks.(ii) Redistribute tasks among PIs (SB, MR) and existing research team to avoid timeline deviation.	Medium	Medium
Departure of key researcher	1	(i) Mandatory documentation/code in shared repositories (Git/Drive). (ii) Ensure 1-month handover period for knowledge transfer.	Low	Medium
Low stakeholder engagement for workshops.	1	(i) Use UPC/ Imperial College London networks to ensure key stakeholders' attendance. (ii) Use hybrid formats to lower barriers; expand scope to master's students.	Medium	Low
Data constraints (Poor quality, gaps, or insufficient granularity).	2,3, 4	(i) Prioritize robust open public databases (PyPSA, historical weather data, etc) as primary sources. (ii) Use synthetic data generation and the proprietary DOME tool (PI1-SB) for error correction.	Medium	Medium
Computational bottlenecks for optimization tools.	3, 4	(i) Prioritize algorithmic efficiency in Python code development. (ii) Utilize the internal CITCEA-UPC HPC cluster for optimization and AI forecasting.	Medium	Low

Table 4 Risks identification. Likelihood, impact and mitigation actions.

3.5. Previous results of the team in the theme of the proposal.

The **research team**, composed by Sara Barja (PI1-SB), Matteo Ranaboldo (PI2-MR) and Samuel Galceran (SG), brings consolidated expertise in energy system planning and multi-criteria EMS

optimization. Both PIs are UPC lecturers. Their relevant experience to supporting the proposal development is detailed below, including the most relevant projects, their roles, and the EAGLE tasks (T.X.X) that are directly benefited. Information about the working team is reported in section 3.6.1.

PI1-SB has participated in **12 competitive** projects focused on enhancing distribution grids through flexibility, EMS, and AI-based services. In 2019, she developed under the **H2020 INVADE** project [62] optimization models for EMS and created flexibility-market mechanisms to respond to flexibility requests from DSOs (T2.1, T4.1, T4.2). Her Master's thesis, based on this work, received Honors and won the Best Thesis Award from the Official College of Industrial Engineers of Catalonia. From 2020-2024, she served as **WP4 leader** and UPC coordinator (T1.1) for the H2020 BD4OPEM project [64], overseeing the development of eighteen AI services for distribution networks. This research culminated in her PhD thesis, which earned the **Excellent Cum Laude** distinction and the UPC **Extraordinary Doctoral Award** for the best thesis in Industrial Engineering. Currently, in the *Horizon Europe ODEON* project [63], she is **Task Leader** of the impact assessment and participates in tasks related with energy communities (T4.2), while also coordinating the UPC team (T1.1). She is the **Scientific Coordinator** of the **Estabanell-UPC Chair**, where she supports the DSO company in driving innovation within its electrical network. She was **co-director** of the **Master Smart Energy** at UPC from 2021-2024 (T1.1). **PI1-SB** has delivered **technology-transfer outcomes** directly relevant to this proposal. In 2022, she **won First Prize** in the "Llavors" category of the *Catalan Energy Efficiency Cluster* for an EMS designed to minimize emissions and cost (T4.2). This work led to **OCEAN Smart Energy** [64], a spin-off initiative supported by accelerators programs such as Barcelona Activa and MWC Collider. The technology was subsequently exhibited at MWC 2024 (4YFN) and GITEX Berlin 2025. Although she is no longer involved in the project, the experience gained in bridging research and market uptake directly supports EAGLE's dissemination and transfer tasks (T1.2). She is the coordinator and lecturer for two courses in the Master's program at UPC, focusing on *Data Science Applied to Electrical Systems* and *Energy Planning*.

PI2-MR brings expertise to the EAGLE project through his **15 years of research and professional experience** in renewable energy development and planning of energy systems. In 2010, he conducted a research stay at DTU, developing forecast models for short-term wind power **forecasts** (T4.1). He developed his PhD thesis, which earned the **Excellent Cum Laude** distinction, between 2011 and 2015 over the development of optimization models for **planning and design of off-grid electrification projects** with renewable energies and storage systems (T3.1, T3.2), and since then has been involved in different research projects over the topic (such as the ENE2010-15509 Spanish project [65]). Between 2015 and 2022, MR has been working in a consultancy company focused on **renewable energy spatial planning and optimization** (T2.2, T3.1), including resource assessment and GIS analysis (T3.2). In 2022 he joined CITCEA-UPC where he has been involved in the *Horizon Europe FLEX4FACT* project [66] as **WP4 Leader**. In this role, he oversaw the development of optimization algorithms to support industrial management enabling the provision of **flexibility services** (T4.2). During last years, he also collaborated in a research line on energy communities (OptIAComE Spanish project [67]), in particular developing **multicriteria optimization** methods (T3.2) for their design and operation. He is giving courses on *industrial energy efficiency*, *energy community managements*, *wind energy* and *electrotechnics*.

SG is an associate professor at UPC, with a remarkable experience in power systems analysis, storage technology design and renewable integration in the electric grid. He has participated in over 29 R&D competitive projects over his career. His recent work targets RES integration for microgrids and controllers for grid-forming. SG will contribute to tasks 2.2, T2.3 and T3.1, besides supporting in project management and coordination (WP1) relying on his previous experience in research project. All components of the research team belong to the group Centre d'Innovació Tecnològica en Convertidors Estàtics i Accionaments (CITCEA-UPC).

3.6. Human, material and equipment resources available for the execution of the Project.

3.6.1 Human resources

The involved project team is as described in Table 5, indicating the background and expertise and the most relevant projects in which they participated.

	Name	Ref	Background and expertise	Projects participation
Research team	Sara Barja PI1	SB	Project coordination. Energy management and demand response. AI expertise and multi-criteria optimization.	ODEON, INVADE, BD4OPEM, OMEGA-X [68], MERIDIAN [69], ATLAS [70], DESIRES [71], CPP2022-009696 [72]
	Matteo Ranaboldo PI2	MR	Project coordination. Energy systems planning, RESS design and industrial demand response. Multi-criteria optimization and GIS analysis.	FLEX4FACT, OptIComE, DESIRES, CPP2022-009696,
	Samuel Garcelan	SG	Project coordination. Power systems analysis, RES integration and storage technology	FLEX4FACT, DESIRES, AGISTIN [73], CPP2022-009696
	Antonio Saldaña	AS	Long-term energy planning, GIS analysis for power systems, grid capacity	ATLAS, OMEGA-X, BD4OPEM
Working team	Adriano Caprara	AC	Flexibility models and markets, data-centre flexibility assessment	FLEX4FACT, BD4OPEM
	Paula Gonzalez	PG	Flexibility and energy management, AI-based forecasting models	FLEX4FACT, FEVER [74]
	Fei Teng	FT	Data centres integration in power networks, grid-constrained optimization	
	Technical Engineer	TE	Electrical engineering and optimization profile with background in GIS analysis for power systems planning	

Table 5 EAGLE research team. In *blue*, are the European competitive projects. In *green*, Spanish national funded competitive projects. In *orange*, private-public collaboration projects.

The **working team** provides expertise in several key areas of the project. Specifically, AS is a postdoc researcher with expertise on optimization and GIS analysis for power systems planning [75], while AC and PG are PhD students focusing on forecast algorithms, data-centre flexibility assessment, and flexibility services and markets [53], [76]. As shown in Table 5, the Research and Working team has actively collaborated on national and international projects over the past few years. These collaborations encompass related fields, including energy system planning, renewable energy system design and grid integration, storage technology, forecasting algorithms, and energy management. Furthermore, the project incorporates the expertise of an international collaborator, FT, who will join the working team. FT, a reader professor in Intelligent Energy Systems at Imperial College London, will contribute to EAGLE by leveraging his expertise in grid-constrained optimization, grid flexibility services and integration of data centres in the power network [77] [78]. His research and background will provide a solid support in defining the assessment scenarios for both the grid-resilient planning and demand-side flexibility tools, ensuring the impact of the project developments. The project will require a **Technical Engineer (TE)** member, with a contract duration of 1.5 years. She/he will have experience in operational research and energy planning, programming and GIS skills to develop optimization algorithms and the open-source interface. Her/his background and gained knowledge will enable her/him to identify and develop the algorithms required for OptiSize and SiteOptimizer, thereby being primarily involved in tasks T3.1 and T3.2. Adequate training will be provided to complement and expand the aforementioned skills. The hiring will be aligned with the “UPC gender equality plan”.

3.6.2 Material and equipment resources

Access to bibliographic resources, scientific journals, and databases is provided through the UPC Library Service. This includes, among others, reference materials in the fields of energy systems and operational research techniques. Software equipment: Python will be the language used for developing all algorithms and the GIS interface. In addition, programs that enable the simulation of electrical power systems to validate the distribution grid power flows performed in Task 2.2 (such as PSS/E, DigSILENT Power Factory, and MATLAB). In order to develop the optimization algorithms for EMS, open-source libraries (e.g. Pyomo or PULP) are used. Hardware equipment: CITCEA-UPC has built a High-Performance Computing (HPC) cluster, which will be used to train, develop and validate the optimization processes of the SiteOptimizer, the OptiSize and the DC-EMS, and perform the simulations in multiple scenarios (T3.3 and T4.3), in case personal laptop computational capacities are not enough. It enables the handling of massive amounts of data at high speeds with parallel performance. The cluster consists of 1 master node and 3 edge nodes. The 3 servers of the edge nodes

have the following specifications. Processors: Intel(R) Xeon(R) Silver 4314 CPU @ 2.40GHz. Cores: 16, Disk: 1TB, GPU: NVIDIA RTX A5000.

4. EXPECTED IMPACT OF THE RESULTS.

4.1. *Expected impact on the generation of scientific-technical knowledge in the thematic area of the proposal.*

The outcomes of EAGLE are aligned with EU and national objectives as presented in Section 2.3. The project is aligned with the pillars of **energy transition**, **digitalization**, and **power system decarbonization** of PNIEC [8]. This project contributes to the strategic line “Energy transition and decarbonization” of the “Energy and mobility” thematic priority. In particular, the optimal location planning and energy management of green data centres considering environmental, economic and social constraints contribute to the following areas of intervention: “Renewable energy for thermal and electrical use and energy storage”, “Smart grids for advanced renewable energy management” and “energy efficiency”. The impact of the project towards the generation of knowledge in the areas of interest is summarized as follows:

- **Advancing multi-criteria optimization frameworks for data centre optimal location planning.** The EAGLE project’s multi-criteria location planning algorithm (i.e., SiteOptimizer and its GIS interface) will enable transparent, interdisciplinary decision-making by minimizing the environmental and social impacts of data centres while ensuring sustainable grid integration. By supporting scenario analysis, trade-off evaluations, and sensitivity testing, these tools will identify critical parameters (such as cooling demand, renewable resource availability, and grid constraints) that most influence optimal siting decisions, thereby advancing data-driven, climate-resilient infrastructure planning.
- **Bridging long-term planning with short-term energy management.** EAGLE will link long-term data centre siting strategies with short-term energy flexibility mechanisms, creating a novel, dual-timescale approach to infrastructure deployment and operation. EAGLE’s integrated framework will demonstrate how optimal siting decisions (e.g., renewable colocation, grid capacity alignment) can enhance real-time flexibility potential (e.g., demand response, load shifting). This contribution will advance scientific understanding of how spatial and temporal dimensions of data centre operations interact, providing a blueprint for climate-resilient digital infrastructure that aligns with EU energy policies and grid decarbonization goals.
- **Data centres as flexibility services providers.** EAGLE will support the transition of data centres from energy consumers into grid flexibility assets, aligning with the EU’s Demand Response and Renewable Energy Directives. By leveraging real-time on-site storage and cooling system operation, the project will demonstrate how data centres can reduce peak demand, stabilize grid frequency, and support renewable integration. This approach not only lowers operational costs but also enhances grid resilience, providing policymakers with actionable strategies to incentivize data centre participation in flexibility markets and ancillary service programs.

4.2. *Social and economic impact of the expected results.*

The project’s social and economic impact is summarized in the key points below. The SDG that this project contributes to are detailed in Section 2.1.

Social impact

- **Democratisation of planning tools (transparency):** Unlike proprietary software, EAGLE develops open-source GIS interfaces, which empower public administrations and small enterprises to analyse complex techno-economic planning without high entry barriers.
- **Accelerated decarbonization:** The project facilitates the transition to 24/7 carbon-free data centre operations. Through the developed algorithms, EAGLE directly reduces GHG emissions, aligning digital infrastructure growth with national and European climate targets.
- **Reduction of grid congestion:** The project relieves bottlenecks in the power network, maximising the on-site renewable and storage generation. Reducing congestion lowers systemic energy costs and improves the power system efficiency.
- **Increment of RES generation:** reducing generation from fossil fuel assets and dependence on other energy sources (e.g. nuclear energy).

Economic impact

- **Avoidance of grid reinforcement costs:** By identifying locations with high hosting capacity, renewables potential, and utilizing demand-side flexibility, the project reduces the need for capital-intensive grid reinforcements in transmission lines and substations.
- **Protection of DC capital cost:** By avoiding site locations classified as high climate risk, the SiteOptimizer significantly enhances the security and longevity of DC investments.
- **New revenue from flexibility markets:** DC-EMS enables data centres to monetize ancillary services, while fulfilling the regulation that encourages DC with a power demand above 500 kW to provide demand-side flexibility.
- **Local economic development:** Targeted DC deployments in renewable-rich regions will attract private investments and create high-tech jobs in construction, O&M, and energy services.

4.3. *Plan for scientific communication and internationalization of the results (indicate the forecast of open access publications).*

The dissemination and internationalization strategy guarantees the effective communication of the project's progress and results on both national and international levels. To achieve this, the following actions are planned:

- **Scientific publications and conferences:**
 - **Journal papers:** 4 open-access papers in high-impact journals (IEEE Transactions, Elsevier...)
 - **Conference papers:** 4 conference papers to lead power systems conferences such as IEEE PES Innovative Smart Grid Technologies (ISGT), CIRED International Conference on Electricity Distribution... The publications will be shared in Open Access.
- **Internationalization:** Attendance at international congresses will be prioritized to disseminate results and foster synergies that expand the project's scope. Additionally, Collaboration with Imperial College London will enable strategic knowledge exchange and visibility within the UK.
- To collaborate with **STEAM-UPC** events, which empower women's visibility in science

4.4. *Plan for dissemination of the results to the most relevant groups for the theme of the project and to society in general.*

Specific actions for the most relevant stakeholders include:

- To hold **3 workshops/webinars. Hybrid format** (online and face-to-face event) to maximize impact:
 - Audience target: *Industry professionals*. Objective: to take into consideration the vision from the DSOs, government, regulators, power system actors regarding the challenges.
 - Audience target: *Researchers and academia*. Objective: Featuring with Imperial College partner to discuss 'How to boost flexibility in Data centres' inviting both students, academics and industry professionals.
 - Audience target: All audiences. Objective: Final outcomes of the project, main challenges faced, roadmap and main conclusions.
- To organize and participate in the **International Conference on Energy Innovation** (organized by CITCEA-UPC and the Endesa Red Chair yearly, with around 100 attendees), focusing on grid hosting capabilities and the new challenge of how to include data centres in a saturated grid.
- Additionally, the following communication actions to the general public will be performed: Presence on the webpage of the CITCEA-UPC and UPC, social media via LinkedIn. Presence in articles for a broad audience; Press releases (one at the start, one at the end of the project). The main results will be shared with the general public in an accessible language via the UPC Citizen Science portal (<https://en.cienciaciudadana.upc.edu/>), and all scientific outputs will be made openly available through the *UPC Open Science platform* (<https://biblioteca.upc.edu/en/investigadors/ciencia-oberta-upc>).

4.5. *Transfer plan and valorization of results*

EAGLE will deliver three core algorithms SiteOptimizer, OptiSize, and DC-EMS. These tools constitute a unified framework that can be directly leveraged by electricity system stakeholders, government

bodies, policymakers, and energy regulatory entities responsible for long-term planning. Beyond academic research, EAGLE will have a concrete pathway for technology transfer to industry. To demonstrate its industrial relevance, the project has secured three Letters of Support from key stakeholders: a computational data centre (the Barcelona Supercomputing Center), a Distribution System Operator (ESTABANELL), and a power systems modelling and planning company (eRoots). These companies have expressed strong interest in the project and a commitment to monitor and validate its outcomes. The technology transfer potential is further strengthened by the PIs' established ecosystem, which includes two active University-Industry Chairs (Cátedras): the Endesa Chair and the Estabanell Chair, both important DSOs and potential stakeholders of the EAGLE project. These existing frameworks ensure direct and effective channels for dissemination and engagement with key end users. Finally, the project's exploitation pathway extends to the building and industry energy management sector, where companies could adopt the DC-EMS for optimization asset management.

4.6. *Summary's management plan of the planned data*

The proposed system approach will be proofed by the Assessment List for Trustworthy AI (ALTAI). In addition, the fundamental principles of research integrity will be respected. Besides, compliance with, i.e., the General Data Protection Regulation applicable in the EU and the Findable, Accessible, Interoperable and Re-usable (**FAIR**) principles to maximize the effectiveness of research will be undertaken. The project final data will be published in UPPCommons Research Data repository (<https://upcommons.upc.edu/dades>), the UPC institutional repository that follows FAIR and it is offered free of charge by UPC to its personnel. Additionally, other open repositories, such as Zenodo, developed under the European OpenAIRE program, will also be considered. A handle and DOI will be provided for the datasets of the project. The developed software tools will be licensed under the Open Data Commons and Creative Commons BY-NC licenses, allowing non-commercial use, and will be linked to the relevant scientific publications, citing the project name assigned by the State Research Agency. Lastly, the eIna DMP tool (<https://dmp.csuc.cat/>) will be used to develop the Data Management Plan, enabling collaborative and online work. The project will not create, process, manage or store any personal data.

4.7. *Effects of gender inclusion in the content of the proposal*

This project is aligned with the Spanish and EU gender balance strategy, balance in research teams and decision making. Regarding **recruitment**, particularly for the Technical Engineer position, gender-inclusive and non-biased language will be strictly applied in all job announcements. From a social perspective, the project will collaborate with **STEAM-UPC** (<https://aquisteam.upc.edu/ca>), a university-led initiative aimed at attracting female talent to science, technology, engineering, and mathematics. This collaboration supports efforts to challenge gender stereotypes and promote equal opportunities in technical fields. The project would be an example of scientific research led and carried out with significant contributions from female researchers.

5. *JUSTIFICATION OF THE REQUESTED BUDGET*

Table 6 presents the project budget. Personnel costs for the Technical Engineer are calculated according to UPC remuneration tables (available [here](#)).

	Cost	Description
Personnel		
Technical Engineer (TE)	86374.82 €	1.5 year of salary (~45k per year). Engineer with experience in energy planning, optimization and GIS interface development.
Travelling costs		
Conference trip and expenses	4000.00 €	4 confereces. Average cost 1000€ (ISGT Europe, CIRED...)
Working team travelling	1000.00 €	2 trips of Imperial College London to UPC (average cost 500€)
Other costs		
Formative courses	1000.00 €	Spend in topic-related with proposal to improve technical skills
Conference fees	3200.00 €	4 conferences. Conf. Average fee cost 800€ (ISGT Europe, CIRED...)
Publicaciones		
Open access fees	8000.00 €	4 journal open access fess (2000€ per publication). This cost may be reduced through UPC's OA publication agreements.
Inventariable		
DS laptop	1000.00 €	1 working laptop for TL
Aux. material ICT	400.00 €	ICT items like monitors, keywors for the TL
TOTAL	104974.82 €	

Table 6 Budget

References

- [1] International Energy Agency (IEA), "Energy and AI," Licence: CC BY 4.0, Paris, 2025.
- [2] International Energy Agency (IEA), «Grid congestion is posing challenges for energy security and transitions,» March, 2025. [En línea]. Available: <https://www.iea.org/commentaries/grid-congestion-is-posing-challenges-for-energy-security-and-transitions>.
- [3] Data Centre Magazine, "Building the next generation of green data centres," p. 3, 19 October 2025.
- [4] G. Kamiya and P. Bertoldi, "Energy Consumption in Data Centres and Broadband Communication Networks in the EU," *Publications Office of the European Union*, 2024.
- [5] European Commission: Joint Research Centre., "Energy consumption in data centres and broadband communication networks in the EU," Publications Office of the European Union., , 2024.
- [6] International Energy Agency (IEA), "Overcoming energy constraints is key to delivering on Europe's data centre goals," 16 November 2025. [Online]. Available: <https://www.iea.org/commentaries/overcoming-energy-constraints-is-key-to-delivering-on-europe-s-data-centre-goals>.
- [7] European Commission, "The AI Continent Action Plan," Shaping Europe's digital future, EU, Brussels, 2025.
- [8] Ministerio para la Transición Ecológica y el Reto Demográfico, "Plan Nacional Integrado de Energía y Clima 2023-2030 (Update)," Gobierno de España, Madrid, 2024.
- [9] European Commission, "The European Green Deal," European Commission, Brussels, 2019.
- [10] European Parliament and the Council of the European Union, "Directive (EU) 2023/1791 on energy efficiency (recast)," Official Journal of the European Union, Brussels, 2023.
- [11] M. **Ranaboldo** and et.al., "A comprehensive overview of industrial demand response status in Europe," *Renewable and Sustainable Energy Reviews*, vol. 203, p. 114797, 2024.
- [12] S. Guillén-Lambea, R. Abrahão and M. Carvalho, "Life cycle assessment of power generation systems in Spain: Exploring a broader view from a consequential perspective," *Sustainable Production and Consumption*, Elsevier, vol. 38, pp. 28-40, 2023.
- [13] Q. Zhang and Z. Meng, "A survey on data center cooling systems: Technology, power consumption modeling and control strategy optimization," *Journal of Systems Architecture*, vol. 119, p. 102253, 2021.
- [14] Data Centers, «How Climate Change Impacts Data Center Site Selection,» online, p. 1, 5 Sep 2023.
- [15] C. Ramos-Cáceres and M. Sandberg, "Planning data center locations in Swedish municipalities. A comparative case study of Luleå and Stockholm," *Cities*, vol. 150, no. 105063, 2024.
- [16] F. H. M. Liu, K. P. Y. Lai, B. Seah and W. T. L. Chow, "Decarbonising digital infrastructure and urban sustainability in the case of data centres," *Nature. Urban sustainability*, no. 15, 2025.
- [17] ENTSOE, "Transparency platform," 1 12 2025. [Online]. Available: <https://transparency.entsoe.eu/>.
- [18] DC Blox Resource Library , "Edge Data Center Deployment Guide," 2025. [Online]. Available: www.dcblox.com/resource-library/.
- [19] S&P Global, "451 Research Reports: Technology Data, Research, & Advisory," 2025. [Online]. Available: www.spglobal.com/market-intelligence/en/solutions/451-research.
- [20] Python for Power System Analysis (PyPSA), "Capacity expansion planning," 2025. [Online]. Available: www.pypsa.org.
- [21] M. T. Takci, M. Qadrdan, J. Summers and J. Gustafsson, "Data centres as a source of flexibility for power systems," *Energy Reports*, vol. 13, pp. 3661-3671, 2025.
- [22] C. Hale, "Many top data center locations could be at risk of climate change," *TechRadar*, 14 Jul 2025.
- [23] Y. Zhang and J. Liu, "Prediction of Overall Energy Consumption of Data Centers in Different Locations," *Sensors*, 2022.
- [24] OpenDCIM, "Open source data center infrastructure management," 2025. [Online].
- [25] Netbox, "Infraestructure resource modeling," 2025. [Online]. Available: <https://github.com/netbox-community/netbox>.
- [26] Open Energy Transition, "Transforming Energy Planning with Open Source and Open Data," 2025. [Online]. Available: www.openenergytransition.org.
- [27] S. Barja-Martinez, F. Rücker, M. Aragüés-Peñalba, R. Villafafila-Robles and P. Lloret-Gallego, "A Novel Hybrid Home Energy Management System Considering Electricity Cost and Greenhouse Gas Emissions Minimization," *IEEE Transactions on Industry Applications*, vol. 57, pp. 2782 - 2790, 2021.
- [28] Ministerio para la Transición Ecológica y el Reto (MITECO), "Sistema Nacional de Cartografía de Zonas (SNCZI)," Nov, Madrid, 2025.
- [29] European Commission, "PVGIS (Photovoltaic Geographical Information System)," 2025. [Online]. Available: https://re.jrc.ec.europa.eu/pvg_tools/es/.
- [30] Energy data info, "Global Wind Atlas," 2025. [Online]. Available: <https://globalwindatlas.info/en/>.
- [31] PyPSA, "PyPSA-Eur: A sector-coupled open optimisation model of the European energy system," 2025. [Online]. Available: <https://pypsa-eur.readthedocs.io/en/latest/>.
- [32] B. Xiong, D. Fioriti, F. Neumann, I. Riepin and T. Brown, "Modelling the high-voltage grid using open data for Europe and beyond," *Nature. Scientific Data*, no. 277, 2025.
- [33] Destination Earth (European union), "DestinE Platform. A gateway to a sustainable future," [Online]. Available: <https://platform.destine.eu/>.
- [34] M. **Ranaboldo**, A. Garcia-Villoria and et.al., "A meta-heuristic method to design off-grid community electrification projects with renewable energies," *Energy*, vol. 93, pp. 2467-2482, 2015.
- [35] M. **Ranaboldo**, Design of off-grid renewable energy community electrification projects : analysis of micro-scale resource variations and development of optimization methods, Barcelona: Ph.D. Thesis, 2015.
- [36] B. Domenech, M. **Ranaboldo** and e. al., "Local and regional microgrid models to optimise the design of isolated electrification projects," *Renewable Energy*, vol. 19, pp. 795-808, 2018.
- [37] T. Bhattacharya, M. Rahgouy and et.al., "Capping carbon emission from green data centers," *International Journal of Energy and Environmental Engineering*, vol. 14, no. 4, 2023.
- [38] The Guardian, "Google's emissions climb nearly 50% in five years due to AI energy demand," 2 Jul 2024. [Online]. Available: <https://www.theguardian.com/technology/article/2024/jul/02/google-ai-emissions>.
- [39] F. Wang and V. Nian, "Do 'green' data centres really have zero CO2 emissions?," *Sustainable Energy Technologies and Assessments*, vol. 53, no. 102769, 2022.
- [40] Latitude Media, "Google expands demand response to target machine learning workloads," 4 Aug 2025. [Online]. Available: <https://www.latitudemedia.com/news/google-expands-demand-response-to-target-machine-learning-workloads/>.
- [41] Ultrus, "HOMER Software. Hybrid Optimization of Multiple Energy Resources," 2025. [Online]. Available: <https://www.homerenergy.com/>.
- [42] NREL, "REopt web tool," 2025. [Online]. Available: <https://reopt.nrel.gov/tool>.
- [43] S. Barja-Martinez and e. I Munné-Collado, "A novel home energy management system environmental-based with LCA minimization," in *2020 IEEE Industrial and Commercial Power Systems Europe (IEEEIC/ICPS Europe)*, Madrid, Spain, 2020.
- [44] M. **Ranaboldo** and et.al., "Renewable energy projects to electrify rural communities in Cape Verde," *Applied Energy*, vol. 118, pp. 280-291, 2014.
- [45] M. **Ranaboldo**, B. Domenech and e. al., "Off-grid community electrification projects based on wind and solar energies: A case study in Nicaragua," *Solar Energy*, vol. 117, pp. 268-281, 2015.
- [46] H. Du and X. Zhou, "A new framework for evaluating and enhancing the performance of district heating systems integrated with data centres

- using short-term thermal energy storage," *Energy*, vol. 319, no. 134934, 2025.
- [47] ASHRAE Technical Committee 9.9, "Data Center Power Equipment Thermal Guidelines and Best Practices," ASHRAE, 2016.
- [48] Massachusetts Institute of Technology (MIT), "Flexible data centers can reduce costs — if not emissions," 20 Oct 2025.
- [49] P. Fisco, M. **Ranaboldo** and et.al., "Sustainable optimal production scheduling methodology under implicit demand response and carbon policy: A manufacturing industry case study," *Applied Energy*, vol. 384, no. 125403, 2025.
- [50] E. Manafi, M. **Ranaboldo** and et.al., "A self-learning whale optimization algorithm based on reinforcement learning for a dual-resource flexible job shop scheduling problem," *Applied Soft Computing*, vol. 180, no. 113436, 2025.
- [51] Horizon Europe, "FLEX4FACT Horizon Europe. Industrial Flexibility Platform for sustainable factories," 2024. [Online]. Available: <https://flex4fact.eu/>.
- [52] S. **Barja-Martinez**, Energy management systems for smart homes and local energy communities based on optimization and artificial intelligence techniques, Barcelona: Phd. Thesis, 2023.
- [53] A. Caprara, S. **Barja-Martinez** and et.al., "Flexibility Service for Balance Responsible Parties' Industrial Customers: A Day-Ahead Cost Optimization Approach Using Price Forecasting," *IEEE Access*, vol. 13, 2025.
- [54] M. **Ranaboldo** and et.al., "Implementation of a Model Output Statistics based on meteorological variable screening for short-term wind power forecast," *Wind Energy*, vol. 16, pp. 811-826, 2013.
- [55] S. Naganandhini, C. Sundar, R. Sathya and G. Venkatesan, "Towards Energy-Efficient Data Centres: A Comprehensive Analysis of Cooling Strategies for Maximizing Efficiency and Sustainability," Chennai, India, 2023.
- [56] Alibaba, "Alibaba data centre data," [Online]. Available: <https://github.com/alibaba/clusterdata>.
- [57] "Awesome Datasets About Datacenter (ADAD)," [Online]. Available: <https://github.com/bean-zhang/awesome-datasets-about-datacenter/tree/master>. [Accessed November 2025].
- [58] OpenStreetMap, "GridKit: European and North-American network models for high-voltage power grid," [Online]. Available: <https://zenodo.org/records/47317>.
- [59] C. ADAPT, "European Climate Risk and Adaptation Platform," [Online]. Available: <https://climate-adapt.eea.europa.eu/en>.
- [60] E. Commission, "European Flood Risk Areas Viewer," [Online]. Available: <https://discomap.eea.europa.eu/floodsviwer/>.
- [61] E. Commission, "DestinE platform: develop a highly accurate digital model of the Earth to monitor and simulate natural phenomena," [Online]. Available: <https://platform.destine.eu/>.
- [62] INVADE H2020 (2017–2020) GA 731148, "Integrated electric vehicles and batteries to empower distributed and centralised storage in distribution grids & markets," [Online]. Available: <https://h2020invade.eu/>.
- [63] ODEON Horizon Europe (2024-2027) GA 101136128,, "federated data and intelligence Orchestration & sharing for the Digital Energy transition," [Online]. Available: <https://odeonproject.eu/>.
- [64] OCEAN Smart energy, "Energy management system for minimizing emissions and cost," [Online]. Available: <https://oceansmartenergy.com/>.
- [65] E.-I. (2011-2014), "Electrificación rural con energía eólica y solar," [Online].
- [66] FLEX4FACT Horizon Europe (2022-2026) GA 101058657, "Industrial Cluster FLEXibility platform for sustainable FACTories to reduce CO2 emissions and to enable the Energy Transition," [Online]. Available: <https://flex4fact.eu/>.
- [67] OptiACoME PID2023-151210OB-I00 (2024-2027), , "Multicriteria optimization and artificial intelligence for the design and the operation of energy communities," [Online].
- [68] OMEGA-X Horizon Europe (2022-2025) GA 101069287, , "Orchestrating an interoperable sovereign federated Multi-vector Energy data space built on open standards and ready for GAia-X," [Online]. Available: <https://omega-x.eu/>.
- [69] MERIDIAN TED2021-131753B-I00 (2022-2025), , "Gestión flexible de la red de distribución para la máxima descarbonización mediante inteligencia artificial," [Online].
- [70] ATLAS PID2021-128101OB-I00 (2022-2025), , "Herramientas analíticas para sistemas eléctricos. Digitalización mediante nuevos métodos y toolboxes de analítica de datos para redes seguras, renovables y flexibles," [Online].
- [71] DESIRES PID2023-151674OA-I00 (2024-2027), , "Developing Enhanced Synthetic Inertia strategies for electricity networks with high ratios of Renewable Energy Sources," [Online].
- [72] CPP2022-009696 (2023-2026), , "Experimental development of new parallel methods in power supply systems," [Online].
- [73] AGISTIN Horizon Europe (2022-2026) GA 101096197, , "Advanced Grid Interfaces for innovative STorage Integration," [Online]. Available: <https://agistin.eu/>.
- [74] FEVER H2020 (2020-2024). GA 864537, "Flexible Energy Production, Demand and Storage-based Virtual Power Plants for Electricity Markets and Resilient DSO Operation," [Online]. Available: <https://fever-h2020.eu/>.
- [75] e. Saldaña-Gonzalez, "Distribution network planning method: Integration of a recurrent neural network model for the prediction of scenarios," *Electric Power Systems Research*, vol. 229, no. 110125, 2024.
- [76] A. Caprara, P. González-Font-de-Rubinat and Bullich, "Energy optimization of an industrial cluster: a cooperative coalition approach," in *IET Conference Proceedings*, vol. 2024, núm. 5, p. 719-722., 2024.
- [77] Y. Sun, F. Teng and e. al., "Learning-Enabled Adaptive Power Capping Scheme for Cloud Data Centers," *IEEE Transactions on Smart Grid*, vol. 16, pp. 4755 - 4767.
- [78] J. Liu, F. Teng and F. Y. Hou, "Synergising Hierarchical Data Centers and Power Networks: A Privacy-Preserving Approach," *IEEE Transactions on Smart Grid*, vol. 16, pp. 5083-5098, Nov 2025.
- [79] American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE), "ANSI/ASHRAE/IES Standard 90.1-2022: Energy standard for buildings except low-rise residential buildings," 2022.
- [80] V. Gadelha and e. al, "Rural electrification planning based on graph theory and geospatial data: A realistic topology oriented approach," *Sustainable Energy, Grids and Networks*, vol. 28, 2021.
- [81] BD4OPEM H2020 (2020-2024) GA 872525, "Big Data for OPen innovation Energy Marketplace," [Online]. Available: <https://bd4opem.eu/>.
- [82] L. Y. B. S. F. T. F. Lei, "Large Scale Data Centers Simulation Based on Baseline Test Model," in *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, Vancouver, BC, Canada, 2018.